

Environmental Chemistry

FE Environmental — Day 9

Study set • 2026-06-25 • 20 questions • every numeric answer recomputed in Python

Handbook fidelity. Handbook formulas confirmed against NCEES FE Reference Handbook v10.6: K_{sp} definition and solubility product equilibrium (Chemistry section); first-order decay $C_t = C_0 \cdot e^{(-kt)}$ and half-life $t_{1/2} = 0.693/k$ (Mathematics — Differential Equations; also Reactor Design table p.331); zero-order integrated rate law $[A]_t = [A]_0 - k \cdot t$ (Reactor Design table p.331); dimensionless Henry's Law constant $H_{cc} = C_{gas}/C_{liq}$ (Environmental Engineering section); retardation factor $R = 1 + (\rho_b/n_e) \cdot K_d$, $K_d = K_{oc} \cdot f_{oc}$ (Environmental Engineering p.330); bioconcentration factor $BCF = C_{organism}/C_{water}$ (Environmental Engineering p.328–329). Equivalent weight table ($EW(HCO_3^-)=61$, $EW(CaCO_3)=50$) in Handbook Common Radicals table p.348 — alkalinity-as- $CaCO_3$ conversion derivable from EW concept (fundamental). NOT in Handbook v10.6: Henderson-Hasselbalch $pH = pK_a + \log([A^-]/[HA])$ — formula supplied in Q03 stem (tagged in-stem, closed-book solvable); strong-acid/base pH ($pH = -\log[H^+]$, $pOH = -\log[OH^-]$, $pH+pOH=14$); ThOD stoichiometry (balanced combustion equation); zero-order half-life (derived in Q10 stem, in-stem); K_{ow} bioaccumulation threshold (regulatory, not in Handbook); degree of unsaturation formula; organic functional-group identification; PCE/TCE structural classification; contaminant velocity $v_c = v_s/R$ (derived from R definition).

Q1. [D9-Q01 • MCQ • pH of a strong acid (HCl)]

A laboratory sample of hydrochloric acid has a concentration of 2.5×10^{-3} mol/L. HCl is a strong acid that dissociates completely in water. The pH of this solution at 25°C is most nearly:

Relevant: $pH = -\log_{10}[H^+]$ (strong acid: $[H^+] = C_{acid}$)

- (A) 1.30
- (B) 2.30
- (C) **2.60 ✓**
- (D) 11.40

Answer: (C)

$[H^+] = 2.5 \times 10^{-3}$ mol/L. $pH = -\log_{10}(2.5 \times 10^{-3}) = -(\log_{10} 2.5 + \log_{10} 10^{-3}) = -(0.398 - 3) = 2.60$. Option (A) 1.30: uses $[H^+] = 0.050$ M (a 20-fold error — likely misplacing the decimal). Option (B) 2.30: uses $[H^+] = 5.0 \times 10^{-3}$ M (factor-of-2 error; $-\log(5 \times 10^{-3}) = 2.30$). Option (D) 11.40: computes pOH correctly ($pOH = 2.60$) but then takes $pH = 14 - 2.60$ as if the solution were basic — acid/base confusion. For a strong acid there is no pOH route; pH is read directly from $[H^+]$.

Handbook: N/A — fundamental ($pH = -\log[H^+]$, strong-acid complete dissociation) • Ctrl-F: pH; Acid; Dissociation

Q2. [D9-Q02 • MCQ • pH of a strong base (NaOH) via pOH]

A dilute NaOH solution has a concentration of 1.0×10^{-3} mol/L. NaOH dissociates completely, producing one OH^- per formula unit. Using $K_w = 1.0 \times 10^{-14}$ at 25°C , the pH of this solution is:

Relevant: $K_w = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ $\text{pOH} = -\log_{10}[\text{OH}^-]$, $\text{pH} + \text{pOH} = 14$

- (A) 3.0
- (B) 4.0
- (C) **11.0** ✓
- (D) 13.0

Answer: (C)

$[\text{OH}^-] = 1.0 \times 10^{-3}$ mol/L. $\text{pOH} = -\log_{10}(1.0 \times 10^{-3}) = 3.0$. $\text{pH} = 14 - 3.0 = 11.0$. Option (A) 3.0: reports pOH as pH (forgetting to subtract from 14). Option (B) 4.0: uses $[\text{OH}^-] = 1.0 \times 10^{-4}$ M (10-fold dilution error). Option (D) 13.0: uses $[\text{OH}^-] = 0.10$ M (100-fold concentration error). At pH 11, $[\text{H}^+] = 10^{-11}$ M — verify: $K_w = 10^{-11} \times 10^{-3} = 10^{-14}$ ✓.

Handbook: N/A — fundamental (K_w , pOH, $\text{pH} + \text{pOH} = 14$) • Ctrl-F: pOH; K_w ; Strong Base; NaOH

Q3. [D9-Q03 • MCQ • Henderson-Hasselbalch equation — buffer pH]

A buffer solution contains 0.10 mol/L of sodium acetate (CH_3COO^-) and 0.05 mol/L of acetic acid (CH_3COOH). The pK_a of acetic acid is 4.74. Using the Henderson-Hasselbalch equation, the pH of this buffer is most nearly:

Relevant: $\text{pH} = \text{pK}_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$

- (A) 4.44 (inverted ratio)
- (B) 4.74 (pK_a only — ratio term omitted)
- (C) **5.04 (correct)** ✓
- (D) 6.74 (ratio added instead of log of ratio)

Answer: (C)

$\text{pH} = 4.74 + \log_{10}(0.10/0.05) = 4.74 + \log_{10}(2.0) = 4.74 + 0.301 = 5.04$. Option (A): inverts the ratio — $\log(0.05/0.10) = \log(0.5) = -0.301$, giving $4.74 - 0.30 = 4.44$. Option (B): ignores the log term entirely; $\text{pH} = \text{pK}_a$ only when $[\text{A}^-] = [\text{HA}]$ (ratio = 1, $\log = 0$). Option (D): adds the ratio directly instead of taking its log: $4.74 + 2.0 = 6.74$ — the log is dimensionless and here equals 0.30, not 2.0. Buffers resist pH change most effectively within ± 1 unit of pK_a ; at pH 5.04 the acetate buffer is in its effective range.

Handbook: N/A — Henderson-Hasselbalch is NOT in FE Reference Handbook v10.6; formula provided in question stem (in-stem, closed-book solvable) • Ctrl-F: Henderson-Hasselbalch; Buffer; pK_a ; Acetic Acid

Q4. [D9-Q04 • Numeric • Alkalinity as CaCO_3 from bicarbonate concentration]

A groundwater sample is analyzed and found to contain 183 mg/L of bicarbonate (HCO_3^-) as the sole contributor to alkalinity. Using equivalent weights $\text{EW}(\text{HCO}_3^-) = 61 \text{ g/eq}$ and $\text{EW}(\text{CaCO}_3) = 50 \text{ g/eq}$ (from the Handbook Common Radicals table), express the total alkalinity in mg/L as CaCO_3 .

$$\text{Relevant: Alkalinity (mg/L as CaCO}_3) = C_{\text{HCO}_3^-} (\text{mg/L}) \times \frac{\text{EW}_{\text{CaCO}_3}}{\text{EW}_{\text{HCO}_3^-}} = C \times \frac{50}{61}$$

Numeric entry.

Answer: 150 mg/L as CaCO_3

Alkalinity = $183 \times (50/61) = 183 \times 0.8197 = 150.0 \text{ mg/L as CaCO}_3$. The factor 50/61 converts from mg/L as HCO_3^- to mg/L as CaCO_3 — the universal water-chemistry reporting basis. $\text{EW}(\text{CaCO}_3) = 100/2 = 50 \text{ g/eq}$ (divalent); $\text{EW}(\text{HCO}_3^-) = 61/1 = 61 \text{ g/eq}$ (monovalent). Alternatively: $183 \text{ mg/L} \div 61 \text{ g/eq} = 3.00 \text{ meq/L}$; $3.00 \text{ meq/L} \times 50 \text{ mg/meq} = 150 \text{ mg/L as CaCO}_3$ (same result). If carbonate (CO_3^{2-}) were also present, its contribution would be $C_{\text{CO}_3^{2-}} \times (50/30)$ added to the HCO_3^- term (EW of $\text{CO}_3^{2-} = 60/2 = 30 \text{ g/eq}$). Common error: dividing instead of multiplying the EW ratio gives $183 \times (61/50) = 223 \text{ mg/L as CaCO}_3$ — an impossible result since alkalinity as CaCO_3 is mg/L of HCO_3^- (HCO_3^- is heavier per equivalent than CaCO_3).

Handbook: Water & Wastewater — Common Radicals table ($\text{EW}(\text{HCO}_3^-)=61$, $\text{EW}(\text{CaCO}_3)=50$, p.348); EW-based conversion is standard analytical chemistry • Ctrl-F: Alkalinity; Bicarbonate; CaCO_3 ; Equivalent Weight

Q5. [D9-Q05 • MCQ • Theoretical oxygen demand (ThOD) — acetone]

Acetone ($\text{C}_3\text{H}_6\text{O}$, MW = 58 g/mol) is a common industrial solvent found in wastewater. Its complete aerobic oxidation follows: $\text{C}_3\text{H}_6\text{O} + 4\text{O}_2 \rightarrow 3\text{CO}_2 + 3\text{H}_2\text{O}$. A wastewater sample contains 80 mg/L of acetone. The ThOD of this sample is most nearly:

$$\text{Relevant: } ThOD = C_{\text{compound}} (\text{mg/L}) \times \frac{\nu MW_{\text{O}_2}}{MW_{\text{compound}}} \quad \nu = \text{moles O}_2 \text{ consumed per mole of compound}$$

- (A) 128 mg/L O_2
- (B) 155 mg/L O_2
- (C) **177 mg/L O_2 ✓**
- (D) 221 mg/L O_2

Answer: (C)

From the balanced equation, $\nu = 4 \text{ mol O}_2/\text{mol acetone}$. $ThOD = 80 \times (4 \times 32/58) = 80 \times (128/58) = 80 \times 2.207 = 176.6 \approx 177 \text{ mg/L O}_2$. Verify the balance: C: $3 = 3\checkmark$; H: $6 = 6$ (in $3\text{H}_2\text{O}$) $\checkmark$; O: $(1 + 8) = 9 = (6 + 3)\checkmark$. Option (A) 128: directly uses the O_2 mass per mole ($4 \times 32 = 128 \text{ g/mol}$) as if it were a concentration — dimensionally incorrect; the ratio 128/58 must be applied. Option (B) 155: substitutes $\text{MW}(\text{N}_2) = 28 \text{ g/mol}$ for $\text{MW}(\text{O}_2) = 32 \text{ g/mol}$ (a common error when working near nitrogen): $80 \times 4 \times 28/58 = 154.5 \approx 155$. Option (D) 221: assumes 5 mol O_2 instead of 4: $80 \times 5 \times 32/58 = 220.7 \approx 221$. The existing oxygen atom in acetone reduces the external O_2 demand relative to propane (C_3H_8 needs 5 mol O_2), hence 4 mol rather than 5.

Handbook: N/A — fundamental stoichiometry (balanced combustion equation) • Ctrl-F: Theoretical Oxygen Demand; ThOD; COD; Stoichiometry

Q6. [D9-Q06 • Numeric • Solubility product (Ksp) — equilibrium Ca^{2+} concentration]

The solubility product of calcite (CaCO_3) at 25°C is $K_{sp} = 3.3 \times 10^{-9}$. In a closed system containing only CaCO_3 and pure water at equilibrium, and assuming no other calcium or carbonate sources, determine the equilibrium Ca^{2+} concentration in mg/L. Use $\text{MW}(\text{Ca}) = 40.1 \text{ g/mol}$.

$$\text{Relevant: } \text{CaCO}_3(s) \rightleftharpoons \text{Ca}^{2+} + \text{CO}_3^{2-} \quad K_{sp} = [\text{Ca}^{2+}][\text{CO}_3^{2-}] = x^2 \quad \Rightarrow \quad x = \sqrt{K_{sp}}$$

Numeric entry.

Answer: $\approx 2.3 \text{ mg/L as } \text{Ca}^{2+}$

Let $x = [\text{Ca}^{2+}] = [\text{CO}_3^{2-}]$ (equal stoichiometry). $x = \sqrt{3.3 \times 10^{-9}} = 5.74 \times 10^{-5} \text{ mol/L}$. $[\text{Ca}^{2+}] = 5.74 \times 10^{-5} \text{ mol/L} \times 40.1 \text{ g/mol} \times 1000 \text{ mg/g} = 2.30 \text{ mg/L}$. In mg/L as CaCO_3 : $5.74 \times 10^{-5} \times 100.1 \times 1000 = 5.75 \text{ mg/L as } \text{CaCO}_3$. Common ion effect: if the water already contains $[\text{Ca}^{2+}] = 100 \text{ mg/L}$ (2.49 mmol/L), the equilibrium $[\text{CO}_3^{2-}] = K_{sp}/[\text{Ca}^{2+}] = 3.3 \times 10^{-9}/2.49 \times 10^{-3} = 1.33 \times 10^{-6} \text{ mol/L}$ — far less than the pure-water value; adding Ca^{2+} suppresses carbonate solubility. This is the thermodynamic basis of CaCO_3 precipitation in water softening and scale formation.

Handbook: Chemistry — Solubility Product (Ksp definition and table, NCEES FE Reference Handbook, Chemistry section) • Ctrl-F: Ksp; Solubility Product; CaCO_3 ; Calcium Carbonate

Q7. [D9-Q07 • MCQ • First-order rate constant from half-life]

A volatile organic compound (VOC) in a contaminated aquifer is observed to degrade via first-order natural attenuation. Field monitoring data show that the contaminant concentration decreases by 50% every 12 days. The first-order rate constant k is most nearly:

$$\text{Relevant: } t_{1/2} = \frac{0.693}{k} \quad \Rightarrow \quad k = \frac{0.693}{t_{1/2}}$$

- (A) 0.029 day^{-1}
- (B) **0.058 day^{-1}** ✓
- (C) 0.083 day^{-1}
- (D) 0.116 day^{-1}

Answer: (B)

$k = 0.693/t_{1/2} = 0.693/12 = 0.05775 \approx 0.058 \text{ day}^{-1}$. Verify: $t_{1/2} = 0.693/0.058 = 11.9 \approx 12 \text{ days}$ ✓. Option (A) 0.029: uses $0.693/(2 \times 12) = 0.0289$ — doubles the half-life in the denominator. Option (C) 0.083: uses $k = 1/t_{1/2} = 1/12 = 0.0833$ — applies a mean lifetime formula ($\tau = 1/k$) instead of half-life ($t_{1/2} = 0.693/k$); the mean lifetime $\tau = 1/k = 17.3 \text{ d}$ is the time for $e^{-1} \approx 37\%$ remaining, not 50%. Option (D) 0.116: uses $k = 0.693/(t_{1/2}/2) = 2 \times 0.058$ — halves the half-life, which would correspond to $t_{1/2} = 6 \text{ d}$. Note: the 0.693 factor comes from $\ln 2$; at exactly one half-life, $C = C_0 e^{-kt_{1/2}} = C_0/2$, so $kt_{1/2} = \ln 2 = 0.693$.

Handbook: Mathematics — Differential Equations ($t_{1/2} = 0.693/k$ for first-order decay, p.20 or near); confirmed in Reactor Design table (p.331) • Ctrl-F: Half-Life; First Order; Rate Constant; Decay

Q8. [D9-Q08 • Numeric • First-order decay — time to reach cleanup target]

A batch treatability study shows that naphthalene biodegrades via first-order kinetics with $k = 0.15 \text{ day}^{-1}$. The initial concentration is $C_0 = 100 \text{ mg/L}$. Determine the time required (in days) for the concentration to reach the remediation target of 5 mg/L .

$$\text{Relevant: } C_t = C_0 e^{-kt} \Rightarrow t = \frac{\ln(C_0/C_t)}{k}$$

Numeric entry.

Answer: ≈ 20 days

$t = \ln(100/5)/0.15 = \ln(20)/0.15 = 2.996/0.15 = 19.97 \approx 20$ days. Verify: $C = 100 \times e^{-0.15 \times 20} = 100 \times e^{-3.0} = 100 \times 0.0498 = 4.98 \approx 5.0 \text{ mg/L}$ ✓. The ratio $C_0/C_t = 20 = 2^{4.32}$ corresponds to 4.32 half-lives ($t_{1/2} = 0.693/0.15 = 4.62$ days; $4.32 \times 4.62 = 20.0 \text{ d}$ — consistent). Common errors: (i) using $t = (C_0 - C_t)/k$ (zero-order formula): $(100 - 5)/0.15 = 633 \text{ d}$ — orders of magnitude wrong; (ii) forgetting $\ln$ and writing $t = (C_0/C_t)/k = 20/0.15 = 133 \text{ d}$ — dimensionally inconsistent.

Handbook: Environmental Engineering — Reactor Design table, PFR/Batch first-order decay: $C_t = C_0 e^{-k\theta}$ (p.331) • Ctrl-F: First Order; Biodegradation; Cleanup Target; Naphthalene

Q9. [D9-Q09 • MCQ • Concentration remaining after multiple half-lives]

A radioactive tracer used in an aquifer test has an initial activity concentration of $C_0 = 500 \text{ } \mu\text{g/L}$ and a physical half-life of $t_{1/2} = 10$ days. What is the concentration remaining after 20 days?

$$\text{Relevant: } C_t = C_0 \left(\frac{1}{2}\right)^{t/t_{1/2}} = C_0 e^{-kt}, \quad k = \frac{0.693}{t_{1/2}}$$

- (A) $68 \text{ } \mu\text{g/L}$
- (B) **$125 \text{ } \mu\text{g/L}$** ✓
- (C) $250 \text{ } \mu\text{g/L}$
- (D) $375 \text{ } \mu\text{g/L}$

Answer: (B)

$t/t_{1/2} = 20/10 = 2.0$ half-lives. $C = 500 \times (1/2)^2 = 500/4 = 125 \text{ } \mu\text{g/L}$. Each half-life halves the concentration: $500 \xrightarrow{10 \text{ d}} 250 \xrightarrow{10 \text{ d}} 125 \text{ } \mu\text{g/L}$. Option (A) 68: uses $k = 1/t_{1/2}$ (mean-lifetime formula): $500 \times e^{-(1/10) \times 20} = 500 \times e^{-2.0} = 67.7 \approx 68 \text{ } \mu\text{g/L}$ (off by $1/\ln 2 = 1.443$). Option (C) 250: applies only one half-life — misreads $t_{1/2}$ as 20 days or stops after step 1. Option (D) 375: subtracts $C_0/4 = 125$ from C_0 linearly: $500 - 125 = 375$ — a zero-order arithmetic error instead of geometric halving.

Handbook: Mathematics — Differential Equations (first-order decay, half-life, p.20) • Ctrl-F: Half-Life; Decay; Radioactive Tracer; Aquifer

Q10. [D9-Q10 • MCQ • Zero-order reaction — half-life depends on initial concentration]

An oxidant in a chlorine-contact chamber degrades the target pathogen according to zero-order kinetics, with rate constant $k_0 = 5$ mg/L·day. The initial concentration is $C_0 = 60$ mg/L. For zero-order kinetics the integrated rate law is $[A]_t = [A]_0 - k_0 t$. The half-life (time to reach $C_0/2 = 30$ mg/L) is most nearly:

Relevant: $[A]_t = [A]_0 - k_0 t$ (zero-order) $t_{1/2}^{\text{zero}} = \frac{[A]_0}{2 k_0}$

- (A) 0.14 day (applies first-order formula with wrong units for k)
- (B) **6 days (correct zero-order half-life) ✓**
- (C) 12 days (uses C_0/k_0 instead of $C_0/(2k_0)$)
- (D) 30 days (uses $C_0/k_0 \times \text{half}$ — decimal error)

Answer: (B)

At $t_{1/2}$: $[A]_0/2 = [A]_0 - k_0 t_{1/2}$. Solving: $t_{1/2} = [A]_0/(2k_0) = 60/(2 \times 5) = 6$ days. Option (A) 0.14 d: blindly applies the first-order formula $t_{1/2} = 0.693/k$ with $k = 5$ (but 5 has units mg/L·day, not day^{-1} — dimensionally incompatible with first-order). Option (C) 12 d: uses $C_0/k_0 = 60/5 = 12$ — forgets the factor of 2 (this is the time to full depletion, $[A] = 0$, not half-depletion). Option (D) 30 d: arithmetic confusion, perhaps $k_0 = 2$ mg/L·day (misreads 5 as 2) giving $60/4 = 15$ or uses $C_0/k_0 \times (1/2) = 6$ (correct but then reports 30 from another mistake). Critical contrast: for a first-order reaction the half-life is INDEPENDENT of C_0 ; for zero-order, $t_{1/2}^{\text{zero}} \propto C_0$ — doubling the initial concentration doubles the zero-order half-life.

Handbook: Environmental Engineering / Chemistry — Reactor Design table (zero-order integrated rate law $[A] = [A]_0 - k_0 t$, p.331) • Ctrl-F: Zero-Order; Half-Life; Rate Law; Kinetics

Q11. [D9-Q11 • MCQ • Organic functional groups — identifying a carboxylic acid]

Organic compounds are classified by their functional groups, which govern reactivity and fate in the environment. Which of the following molecules contains a carboxylic acid functional group ($-\text{COOH}$)?

- (A) CH_3OH (methanol)
- (B) **CH_3COOH (acetic acid) ✓**
- (C) CH_3NH_2 (methylamine)
- (D) CH_3CHO (acetaldehyde)

Answer: (B)

Option (B) CH_3COOH (acetic acid) contains the carboxylic acid group $-\text{COOH}$, which consists of a carbonyl $\text{C}=\text{O}$ and a hydroxyl $-\text{OH}$ bonded to the same carbon. Carboxylic acids are weak acids ($\text{pK}_a \sim 4\text{--}5$); acetic acid $\text{pK}_a = 4.74$. Option (A) CH_3OH (methanol): contains a hydroxyl $-\text{OH}$ group $\rightarrow$ alcohol (not acidic unless deprotonated at $\text{pH} \gg 15$). Option (C) CH_3NH_2 (methylamine): contains $-\text{NH}_2 \rightarrow$ primary amine (acts as a BASE, $\text{pK}_b \sim 3.4$). Option (D) CH_3CHO (acetaldehyde): contains $-\text{CHO} \rightarrow$ aldehyde (carbonyl at the chain end, no OH); not acidic. Environmental relevance: carboxylic acids such as acetic, propionic, and butyric acids are key intermediates in anaerobic biodegradation; their accumulation signals a stressed digester. Acetic acid is also the primary substrate for methanogenic archaea.

Handbook: N/A — fundamental organic chemistry (functional group identification) • Ctrl-F: Carboxylic Acid; Functional Group; Organic Chemistry

Q12. [D9-Q12 • MCQ • Degree of unsaturation (index of hydrogen deficiency)]

Benzoic acid has the molecular formula $C_7H_6O_2$. Using the degree-of-unsaturation formula (oxygen and sulfur atoms are not counted), how many degrees of unsaturation does benzoic acid contain, and which structural features account for them?

Relevant: $DoU = \frac{2C + 2 - H + N}{2}$ (O and S not counted; each halogen replaces one H)

- (A) DoU = 4 — benzene ring only (3 C=C bonds + 1 ring)
- (B) **DoU = 5 — benzene ring (4) + C=O of carboxyl group (1) ✓**
- (C) DoU = 6 — benzene + two carbonyl groups
- (D) DoU = 3 — three double bonds, no ring counted

Answer: (B)

$DoU = (2 \times 7 + 2 - 6)/2 = (14 + 2 - 6)/2 = 10/2 = 5$. Structural breakdown: the benzene ring contributes 4 degrees (3 C=C double bonds + 1 ring closure); the carboxylic acid $-COOH$ group contributes 1 degree (the C=O carbonyl). Total = 5. Option (A): counts only the benzene ring (DoU=4) and ignores the carboxyl C=O. Option (C): double-counts by assuming both the C=O and the O–H bond each add a degree (only the C=O counts). Option (D): counts 3 C=C bonds but ignores the ring, which adds 1 extra degree over an open-chain structure with the same number of double bonds. Practical use: DoU helps identify compound class — DoU=4 → likely aromatic; DoU ≥ 1 → unsaturated or cyclic; DoU=0 → saturated (alkane or alcohol).

Handbook: N/A — fundamental organic chemistry (degree of unsaturation formula) • Ctrl-F: Degree of Unsaturation; Index of Hydrogen Deficiency; Aromatic

Q13. [D9-Q13 • MCQ • Organic compound classes — identifying a saturated alkane by molecular formula]

Alkanes are saturated hydrocarbons with the general formula C_nH_{2n+2} and zero degrees of unsaturation. Which of the following molecular formulas represents a saturated alkane?

Relevant: Alkane general formula: C_nH_{2n+2} ($n \geq 1$)

- (A) C_6H_6 (DoU = 4 — benzene, aromatic)
- (B) C_4H_8 (DoU = 1 — butene or cyclobutane)
- (C) **C_5H_{12} (DoU = 0 — pentane, saturated alkane) ✓**
- (D) C_3H_4 (DoU = 2 — propyne or allene)

Answer: (C)

For $n = 5$: $C_5H_{2(5)+2} = C_5H_{12} \rightarrow DoU = (2 \times 5 + 2 - 12)/2 = 0$. Zero unsaturation confirms no double bonds or rings — this is pentane. Option (A) C_6H_6 : $DoU = (12 + 2 - 6)/2 = 4 \rightarrow$ benzene (aromatic). Option (B) C_4H_8 : $DoU = (8 + 2 - 8)/2 = 1 \rightarrow$ 1-butene (C=C) or cyclobutane (ring). Option (D) C_3H_4 : $DoU = (6 + 2 - 4)/2 = 2 \rightarrow$ propyne (C≡C counts as DoU=2) or allene (two C=C). Environmental significance: simple alkanes in petroleum are hydrophobic (low water solubility), persist in soil, and are biodegraded aerobically by bacteria that use alkane monooxygenases. BTEX compounds (benzene, toluene, ethylbenzene, xylene) — all aromatic — are priority groundwater pollutants because of their higher water solubility and toxicity relative to straight-chain alkanes.

Handbook: N/A — fundamental organic chemistry (alkane formula, degree of unsaturation) • Ctrl-F: Alkane; Saturated Hydrocarbon; Organic Chemistry; BTEX

Q14. [D9-Q14 • MCQ • Chlorinated solvents — PCE vs TCE molecular structure]

Tetrachloroethylene (PCE, also called perchloroethylene) and trichloroethylene (TCE) are the two most commonly detected chlorinated solvents in contaminated groundwater at CERCLA (Superfund) sites. Which statement correctly describes their molecular formulas and a key structural difference?

- (A) PCE = C_2HCl_3 (one hydrogen); TCE = C_2Cl_4 (no hydrogen) — reversed assignment
- (B) **PCE = C_2Cl_4 (no hydrogen atoms); TCE = C_2HCl_3 (one hydrogen atom)** ✓
- (C) Both PCE and TCE are single-carbon compounds (only one carbon per molecule)
- (D) PCE contains a $C\equiv C$ triple bond; TCE contains a $C=C$ double bond

Answer: (B)

PCE = perCHLOROethylene = C_2Cl_4 (the 'per' prefix means fully halogenated — no remaining C–H bonds). TCE = TRIchloroethylene = C_2HCl_3 (three Cl, one H remaining on the double bond). Both contain a $C=C$ double bond (alkene/chloroethylene structure). Option (A) reverses the assignment. Option (C) is wrong: both have two carbons (ethylene = 2-carbon backbone). Option (D) is wrong: neither has a triple bond; both are chloroethylenes with $C=C$. Environmental context: PCE → TCE → DCE (dichloroethylene) → vinyl chloride (VC) → ethylene is the reductive dechlorination pathway in anaerobic groundwater. VC is a known human carcinogen and is often the regulatory-driver compound at chlorinated-solvent plumes. PCE and TCE are both probable human carcinogens and are denser than water (DNAPLs — dense non-aqueous phase liquids), sinking to the bottom of aquifers.

Handbook: N/A — fundamental environmental organic chemistry (chlorinated solvent identification) • Ctrl-F: PCE; TCE; Chlorinated Solvent; DNAPL; Reductive Dechlorination

Q15. [D9-Q15 • MCQ • Henry's Law (dimensionless) — equilibrium air-phase concentration]

A volatile organic compound has a dimensionless Henry's Law constant $H_{cc} = 0.35$ ($H_{cc} = C_{gas} / C_{liq}$, concentrations in the same units) at 20°C. A contaminated water sample contains 8.0 mg/L of the compound. At equilibrium, the vapor-phase concentration above the water surface is most nearly:

Relevant: $H_{cc} = \frac{C_{gas}}{C_{liq}}$ (dimensionless; same mass/volume units both phases) $C_{gas} = H_{cc} \times C_{liq}$

- (A) **2.8 mg/L (air)** ✓
- (B) 8.35 mg/L (air)
- (C) 22.9 mg/L (air)
- (D) 0.044 mg/L (air)

Answer: (A)

$C_{gas} = H_{cc} \times C_{liq} = 0.35 \times 8.0 = 2.8$ mg/L (air). Option (B) 8.35: adds H_{cc} to C_{liq} as if it were a concentration: $8.0 + 0.35 = 8.35$ (dimensionally incorrect — H_{cc} is dimensionless). Option (C) 22.9: inverts — $C_{liq}/H_{cc} = 8.0/0.35 = 22.9$, treating the formula as $H_{cc} = C_{liq}/C_{gas}$ rather than C_{gas}/C_{liq} . Option (D) 0.044: $H_{cc}/C_{liq} = 0.35/8.0 = 0.044$ (takes a ratio the wrong way). Henry's Law in this form is used to assess whether a compound will volatilize from water ($H_{cc} > 0.01 \rightarrow$ significant volatilization; the compound here, at $H_{cc} = 0.35$, readily partitions into air). Air-stripping towers exploit high H_{cc} to remove VOCs from groundwater.

Handbook: Environmental Engineering — Henry's Law ($H_{cc} = C_{gas}/C_{liq}$, dimensionless form, NCEES FE Reference Handbook Environmental Chemistry section) • Ctrl-F: Henry's Law; H_{cc} ; Volatilization; VOC; Air Stripping

Q16. [D9-Q16 • MCQ • Octanol-water partition coefficient (K_{ow}) — bioaccumulation potential]

The octanol-water partition coefficient K_{ow} ($K_{ow} = C_{octanol} / C_{water}$ at equilibrium) is a key predictor of a compound's tendency to partition into organic matter and lipid tissues. Four organic compounds have the following $\log K_{ow}$ values: compound A (-0.5), compound B (1.8), compound C (3.6), compound D (5.2). Which compound is MOST likely to bioaccumulate in the fatty tissues of aquatic organisms?

Relevant: $K_{ow} = \frac{C_{octanol}}{C_{water}}$ (higher $K_{ow} \rightarrow$ more lipophilic)

- (A) Compound A ($\log K_{ow} = -0.5$) — very hydrophilic, stays in water
- (B) Compound B ($\log K_{ow} = 1.8$) — slightly lipophilic, low bioaccumulation
- (C) Compound C ($\log K_{ow} = 3.6$) — moderately lipophilic
- (D) **Compound D ($\log K_{ow} = 5.2$) — highly lipophilic, greatest bioaccumulation ✓**

Answer: (D)

Compound D ($\log K_{ow} = 5.2$, $K_{ow} = 10^{5.2} = 158,000$) has the highest affinity for lipid (fat) relative to water — it partitions most strongly into biological tissue. General classification: $\log K_{ow} < 1 \rightarrow$ hydrophilic (little tendency to accumulate); $1-3 \rightarrow$ moderately lipophilic; $> 3-4 \rightarrow$ bioaccumulation concern; $> 5 \rightarrow$ persistent bioaccumulative toxic (PBT) status and priority pollutant concern. Compounds A through D represent the range from polar/hydrophilic (A: e.g., glucose-like) to highly lipophilic (D: e.g., DDT, PCB congeners, PFAS). High K_{ow} also means: (i) high $K_{oc} \rightarrow$ strong sorption to soil organic carbon; (ii) low water solubility; (iii) tendency to partition to NAPL phases in contaminated sites. The BCF (bioconcentration factor) correlates with K_{ow} : $\log BCF \approx 0.85 \log K_{ow} - 0.70$ (empirical, supplemental — not in Handbook).

Handbook: Environmental Engineering — K_{ow} table and definition (NCEES FE Reference Handbook, Environmental Chemistry section) • Ctrl-F: Kow; Octanol-Water; Bioaccumulation; Lipophilic; Partition Coefficient

Q17. [D9-Q17 • Numeric • Retardation factor from K_{oc} and soil properties (SI)]

A dissolved organic contaminant in an unconfined aquifer has an organic-carbon partition coefficient $K_{oc} = 200$ mL/g. The aquifer matrix has: organic carbon fraction $f_{oc} = 0.010$, bulk density $\rho_b = 1.6$ g/cm³, and effective porosity $n_e = 0.40$. Calculate the retardation factor R . (Note: mL/g = cm³/g, so units are consistent with ρ_b in g/cm³.)

Relevant: $R = 1 + \frac{\rho_b}{n_e} K_d$, $K_d = K_{oc} f_{oc}$

Numeric entry.

Answer: $R = 9.0$

$K_d = K_{oc} \times f_{oc} = 200 \times 0.010 = 2.0$ mL/g = cm³/g. $R = 1 + (\rho_b/n_e) \times K_d = 1 + (1.6/0.40) \times 2.0 = 1 + 4.0 \times 2.0 = 1 + 8.0 = 9.0$. Interpretation: the contaminant travels at 1/9 the seepage velocity of groundwater — for every 9 m the water moves, the contaminant front advances 1 m. Common errors: (i) forgetting to add 1: $R = 8.0$ (conservative tracer has $R = 1$, not 0). (ii) Using $\rho_b = 1.6$ lb/ft³ with K_d in mL/g — must use consistent units. (iii) Confusing K_{oc} with K_{ow} : $K_{oc} = 0.41 \times K_{ow}$ is an empirical relationship (supplemental, not in Handbook) — if K_{ow} is given, the approximate relationship may be stated in-stem. This is the same Handbook formula used in groundwater remediation (Day 6) now applied from the environmental-chemistry multimedia partitioning perspective.

Handbook: Environmental Engineering — Retardation Factor ($R = 1 + (\rho_b/n_e)K_d$, $K_d = K_{oc} \cdot f_{oc}$, p.330) • Ctrl-F: Retardation Factor; K_{oc} ; K_d ; Organic Carbon; Contaminant Transport

Q18. [D9-Q18 • MCQ • Contaminant transport velocity — retardation factor (USCS)]

An aquifer has a measured groundwater seepage velocity of 1.8 ft/day. A sorbing organic contaminant has a retardation factor $R = 9.0$ (calculated from K_{oc} and soil properties). The velocity at which the center of the contaminant plume advances is most nearly:

Relevant: $v_c = \frac{v_s}{R}$ (contaminant velocity = seepage velocity/ R)

- (A) **0.20 ft/day** ✓
- (B) 0.60 ft/day
- (C) 1.8 ft/day
- (D) 16.2 ft/day

Answer: (A)

$v_c = v_s/R = 1.8/9.0 = 0.20$ ft/day. Option (B) 0.60: divides by $\sqrt{R} = 3$: $1.8/3 = 0.60$ — a square-root error. Option (C) 1.8: ignores retardation ($R = 1$ error) — this is the groundwater seepage velocity, not the contaminant velocity. Option (D) 16.2: multiplies instead of divides: $1.8 \times 9.0 = 16.2$ — applies retardation in the wrong direction. The retardation factor physically means the groundwater moves 9 ft while the contaminant moves 1 ft. In cleanup planning: if a plume source is 1,800 ft from a receptor, groundwater arrives in $1800/1.8 = 1,000$ days while the contaminant plume front arrives in $1800/0.20 = 9,000$ days — demonstrating why natural attenuation timescales for sorbing organics can span decades.

Handbook: N/A — fundamental (contaminant velocity = seepage velocity / R , derived from R definition) • Ctrl-F: Retardation Factor; Contaminant Velocity; Plume; Transport

Q19. [D9-Q19 • MCQ • Air-water equilibrium partition — fraction in each phase]

A sealed remediation chamber contains 3.0 ft³ of contaminated air and 1.5 ft³ of contaminated water. A volatile organic compound is present at a total mass of $M_0 = 45$ mg distributed between the two phases. The dimensionless Henry's constant is $H_{cc} = 0.20$ ($H_{cc} = C_{\text{air}}/C_{\text{water}}$). At equilibrium, the mass of compound residing in the air phase is most nearly: [Hint: write the mass balance $C_w \cdot V_w + H_{cc} \cdot C_w \cdot V_a = M_0$ and solve for C_w , then compute mass in air = $H_{cc} \cdot C_w \cdot V_a$.]

$$\text{Relevant: } C_w V_w + H_{cc} C_w V_a = M_0 \quad \Rightarrow \quad C_w = \frac{M_0}{V_w + H_{cc} V_a} \quad m_{\text{air}} = H_{cc} C_w V_a = M_0 \times \frac{H_{cc} V_a}{V_w + H_{cc} V_a}$$

- (A) 7.5 mg (equal-volume assumption $V_a = V_w$)
- (B) 9.0 mg (takes $H_{cc} \times M_0$ directly, ignoring volume ratio)
- (C) **12.9 mg (correct — accounts for $V_a = 2V_w$) ✓**
- (D) 30.0 mg (ignores H_{cc} , distributes by volume fraction only)

Answer: (C)

$m_{\text{air}} = M_0 \times \frac{H_{cc} V_a}{V_w + H_{cc} V_a} = 45 \times \frac{0.20 \times 3.0}{1.5 + 0.20 \times 3.0} = 45 \times \frac{0.60}{1.5 + 0.60} = 45 \times \frac{0.60}{2.10} = 45 \times 0.2857 = 12.86 \approx 12.9$ mg. Option (A) 7.5: assumes $V_a = V_w$: $45 \times H_{cc}/(1 + H_{cc}) = 45 \times 0.20/1.20 = 7.5$ (valid only for equal volumes). Option (B) 9.0: multiplies $H_{cc} \times M_0 = 0.20 \times 45 = 9.0$ — omits the equilibrium volume distribution. Option (D) 30.0: distributes by volume fraction with no Henry's Law: $V_a/(V_a + V_w) \times M_0 = (3/4.5) \times 45 = 30$ (ignores the chemical preference for water when $H_{cc} < 1$). Mass in water: $45 - 12.9 = 32.1$ mg; check: $C_w = 32.1/1.5 = 21.4$ mg/ft³; $C_a = 12.9/3.0 = 4.30$ mg/ft³; $H_{cc} = 4.30/21.4 = 0.201 \approx 0.20$ ✓.

Handbook: Environmental Engineering — Henry's Law ($H_{cc} = C_{\text{gas}}/C_{\text{liq}}$, dimensionless); mass balance is fundamental • Ctrl-F: Henry's Law; Air-Water Partition; Fugacity; VOC; Phase Distribution

Q20. [D9-Q20 • MCQ • Bioconcentration factor (BCF) — definition and bioaccumulation significance]

A regulatory study samples bluegill sunfish from a river receiving an industrial discharge. The whole-body tissue concentration of a chlorinated compound is measured at 4,200 $\mu\text{g}/\text{kg}$ (wet weight). The compound's concentration in the river water is 0.42 $\mu\text{g}/\text{L}$. Assuming fish tissue density $\approx 1 \text{ kg}/\text{L}$, the bioconcentration factor (BCF) and its environmental significance are:

Relevant: $BCF = \frac{C_{\text{organism}} (\mu\text{g}/\text{kg})}{C_{\text{water}} (\mu\text{g}/\text{L})} \quad [\text{L}/\text{kg}] \quad (\text{same units: } \text{kg} \approx \text{L for tissue})$

- (A) BCF = 0.0001 L/kg — compound preferentially stays in water
- (B) BCF = 1,000 L/kg — borderline bioaccumulation (10 $\times$ error in C ratio)
- (C) **BCF = 10,000 L/kg — significant bioaccumulation; compound exceeds EPA concern threshold ✓**
- (D) BCF = 100,000 L/kg — extreme bioaccumulation (100 $\times$ error)

Answer: (C)

$BCF = 4,200 (\mu\text{g}/\text{kg}) / 0.42 (\mu\text{g}/\text{L}) = 10,000 \text{ L}/\text{kg}$. Option (A): inverts — $0.42/4,200 = 10^{-4}$. Option (B): decimal error (perhaps divides 4,200 by 4.2 instead of 0.42): $4,200/4.2 = 1,000$. Option (D): $4,200/0.042 = 100,000$ (misplaces decimal in water concentration). Interpretation: $BCF = 10,000 > 1,000 \rightarrow$ the EPA classifies compounds with $BCF \geq 1,000$ as bioaccumulative (Tier 2 concern) and $BCF \geq 5,000$ as highly bioaccumulative (potential PBT status). This compound concentrates in fish tissue at 10,000 times the water concentration — consistent with a $\log K_{ow} \approx 4.5\text{--}5$ compound (e.g., a heavier PCB congener). BCF differs from biomagnification factor (BMF), which describes concentration increase up a food chain (predator/prey), and from bioaccumulation factor (BAF), which includes dietary uptake.

Handbook: Environmental Engineering — Bioconcentration Factor $BCF = C_{\text{organism}}/C_{\text{water}}$ (NCEES FE Reference Handbook, Environmental Engineering section, p.328–329) • Ctrl-F: Bioconcentration Factor; BCF; Bioaccumulation; Kow; PBT